

Arash Jalil Khabbazi, MS

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Profile

I work on model-based and model-free control methods, including model predictive control (MPC) and reinforcement learning (RL), and apply machine learning (ML) to make buildings, distributed energy resources, and other artificial intelligence (AI)-driven technologies smarter and more efficient.

Education

- Purdue University, IN, USA** 2023 – Present
PhD in Mechanical Engineering — **GPA:** 3.86/4.0
Minor in Computational Science and Engineering
Advisor: [Kevin J. Kircher](#)
- University of British Columbia, BC, Canada** 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.0/4.0 (94%)
Thesis: [Mixing gaseous hydrogen into natural gas distribution pipelines](#)
Advisor: [Sunny R. Li](#)
- University of Tabriz, EA, Iran** 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+

Publications — Full list on [Google Scholar](#)

Journal articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review,” *Applied Energy*, 2025. [\[DOI\]](#)
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions,” *International Journal of Hydrogen Energy*, 2024. [\[DOI\]](#)

Conference papers

6. **A.J. Khabbazi** and K.J. Kircher, “Small HVAC control demonstrations in larger buildings often overestimate savings,” *under review*.

5. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, “Active current limiting control of residential appliances for breaker panel protection across the US: a parametric study,” *CISBAT*, 2025, pp. 1–6.
4. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?,” *International High Performance Buildings Conference*, 2024, pp. 1–10. [\[DOI\]](#)
3. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions,” *Canadian Society for Mechanical Engineering International Congress*, 2023, pp. 1–6. [\[DOI\]](#)
2. **A. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid,” *Canadian Society for Mechanical Engineering International Congress*, 2022, pp. 1. [\[DOI\]](#)
1. **A. Khabbazi**, R. Li, and J. Quinn, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines,” *International Green Energy Conference (IGEC-XIII)*, 2021, pp. 1. [\[DOI\]](#)

Industry experience

- Applied ML Intern, [Tesla](#), Reno, NV, USA** 05/2025 – 08/2025
- Built supervised ML models (XGBoost, neural nets, SVR, and ensembles) for defect detection and quality assessment.
 - Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing.

- Developed data pipelines in MySQL and engineered features to extract high-signal inputs for machine learning models.
- Collaborated on projects projected to reduce testing costs by over \$1M annually through improved quality assessment.

Student Researcher, [FortisBC](#), *Vancouver, BC, Canada* 05/2021 – 08/2023

- Performed assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H₂Lab).
- Worked with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.

Research experience

PhD Research Assistant, Purdue University, *West Lafayette, IN, USA* 08/2023 – Present

- Supported commissioning of [Herrick Labs](#) by upgrading the building automation system and integrating Modbus.
- Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy).
- Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model.
- Reviewed 100+ publications on MPC and RL in HVAC, leading to a published [review paper](#) in Applied Energy.
- Assisted in developing active current-limiting control for appliances to prevent breaker overloads in electrified homes.

MS Research Assistant, University of British Columbia, *BC, Canada* 05/2021 – 08/2023

- Modeled hydrogen injection into natural gas pipelines with CFD by developing ideal- and real-gas mixture models.
- Produced a [journal paper](#), three conference papers, two awards, and a [master's thesis](#) based on the research findings.

Teaching experience

MS Teaching Assistant, University of British Columbia, *BC, Canada* 05/2021 – 08/2023

- Taught *Engineering Analysis I* (Calculus I) for three semesters with student evaluation ratings above 80%.
- Conducted *Fluid Mechanics II* and *Heat Transfer Applications* labs; prepared materials and supervised experiments.

Skills

- **Programming:** Python, C, MATLAB, R, MySQL, Git, Linux
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, Keras
- **Data & Analytics:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Grafana
- **Modeling & Optimization:** CVX, CVXPY
- **Systems & Integration:** Tridium Niagara, InfluxDB, Modbus, IO-Link, MTConnect

Selected Honors & Awards

- ASHRAE Graduate Student Grant-in-Aid Award 2025
- Best Paper Award at CSME 2023 Conference 2023
- Best Presentation Award at CSME 2022 Conference, Advanced Energy Symposium 2022
- UBC Graduate Scholarship 2022
- UBC Dean's Entrance Scholarship 2021

Conferences

- 8th International High Performance Buildings Conference, *West Lafayette, USA* 2024
- ASHRAE Winter Conference, *Chicago, USA* 2024
- Canadian Society for Mechanical Engineering (CSME) Conference, *Sherbrooke, Canada* 2023
- Canadian Society for Mechanical Engineering (CSME) Conference, *Edmonton, Canada* 2022
- 13th International Green Energy Conference, *Virtual* 2021

Workshops & seminars

- Intelligent Building Operations (IBO) Workshop, *West Lafayette, USA* 2024
Topic: What have we learned from field demonstrations of advanced commercial HVAC control? [\[Recording\]](#)

Academic service

Reviewing:

- *Elsevier: Building and Environment*
 - *8th International High Performance Buildings Conference*, Purdue University
- 09/2025 – Present

05/2024

Selected Courses

Fall 2025	IE691, Reinforcement Learning and Control
Fall 2025	ECE570, Artificial Intelligence
Spring 2025	ME597, Distributed Energy Resources
Spring 2025	STAT511, Statistical Methods
Fall 2024	AAE561, Intro to Convex Optimization
Fall 2024	ME518, Analysis of Thermal Systems
Spring 2024	ME597, Industrial IoT Implementation
Spring 2024	ME500, Advanced Thermodynamics
Fall 2022	Applied Machine Learning